A

Mini Project Report on

Job Recommendation and Resume Parsing with NLP and Chatbot

Submitted in partial fulfillment of the requirements for the

degree

Third Year Engineering – Computer Science Engineering (Data Science)

by

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CERTIFICATE

This to certify that the Mini Project report on Job Recommendation and Resume Parsing with NLP and Chatbot has been submitted by Shuban Devanpelli (22107023) , Rishi Bije (22107002) ,Yash Kamble (22107055) and Dhanraj Bacche (22107056) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2024-2025** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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ABSTRACT

This project aims to develop an advanced, AI-driven job recommendation and resume parsing application, that leverages cutting-edge technologies to enhance the job-seeking and recruitment experience. By utilizing Natural Language Processing (NLP), the system will analyze and extract key information from resumes, matching candidates to the most relevant job opportunities based on their skills, experience, and preferences. A Machine Learning-based recommendation engine will predict the best job matches by considering user profiles, job descriptions, and market demand, improving over time through continuous learning. Additionally, an AI-powered chatbot will assist users by providing answer to the queries related to job applications. Through this approach, the project addresses the limitations of traditional job portals by offering a personalized, intelligent, and interactive recruitment experience, streamlining the job search process and improving hiring efficiency.

Chapter 1

Introduction

In today's competitive job market, finding the right job can be a complex and time-consuming process. Traditional job portals often rely on keyword-based matching, which may not always align with a candidate's actual skills or a company's specific requirements. To address these challenges, this project introduces an advanced Job Recommendation and Resume Parsing System that leverages Natural Language Processing (NLP) and an AI-powered Chatbot to enhance the job-seeking and recruitment experience.

This system utilizes NLP algorithms to analyze and extract structured information from resumes, including skills, experience, education, and certifications, ensuring precise and accurate candidate profiling. By leveraging Machine Learning-based job recommendation models, the system intelligently matches candidates with relevant job opportunities based on their qualifications, preferences, and evolving industry trends. The recommendation engine continuously learns from user interactions, job market fluctuations, and employer requirements, ensuring personalized job suggestions.

By integrating cutting-edge AI technologies, this system transforms job search and recruitment into a smarter, faster, and more effective experience. Job seekers benefit from tailored job recommendations, resume enhancements, and career insights, while recruiters experience seamless resume screening, efficient hiring. This innovative approach bridges the gap between job seekers and employers, ensuring a more data-driven, automated, and personalized hiring experience. Ultimately, this system redefines the future of recruitment, making it more accurate, scalable, and user-friendly for all stakeholders

1.1 Purpose

The main purpose of a Job Recommendation and Resume Parsing System is to enhance the job search and recruitment process by providing personalized job recommendations, automated resume screening, and real-time career guidance. By leveraging Natural Language Processing (NLP), Machine Learning (ML), and AI-powered Chatbots, the system helps both job seekers and recruiters streamline hiring, improve matching accuracy, and optimize career development.

For job seekers, the system simplifies job hunting by analyzing their skills, experience, education, and career preferences to generate highly relevant job recommendations. Instead of sifting through thousands of job listings, candidates receive AI-driven, tailored job suggestions based on their qualifications and industry trends. Additionally, the resume parsing functionality ensures that resumes are optimized for Applicant Tracking Systems (ATS), improving job seekers' chances of securing interviews.

Whether it's finding the right job, improving resumes, the system provides smart recommendations and career insights to enhance the overall recruitment experience. The AI chatbot further engages users by answering job-related queries, providing interview tips, and assisting in job applications, ensuring a seamless and interactive job search journey.

1.2 Problem Statement:

Users struggle to find relevant job opportunities because existing job portals rely on keyword-based matching rather than contextual NLP-driven recommendations. Job seekers lack insights into their career progress, required skills, and job market trends due to inadequate data visualization and predictive analytics. The absence of AI-driven chatbots leaves job seekers without personalized career guidance and real-time support. Many existing platforms have complicated interfaces that make job searching and resume submission cumbersome.

1.3 Objectives :

The objective of this project is to recommend suitable jobs to users based on their resume and provide key market insights. It aims to use AI techniques like resume parsing, content-based filtering, and data visualization to guide users in making better career decisions.

1. **To design an intelligent resume parsing tool by implementing Natural Language Processing (NLP) algorithms, enabling efficient extraction and structuring of candidate data.**:- The Job Recommendation System aims to develop an intelligent resume parsing tool using Natural Language Processing (NLP) to automate the extraction and structuring of candidate data. By leveraging NLP techniques like Named Entity Recognition (NER) and semantic analysis, the tool accurately identifies skills, experience,

education, and certifications from resumes, reducing manual screening efforts. The structured data can be integrated into Applicant Tracking Systems (ATS) or job recommendation platforms, enabling faster, more efficient, and data-driven hiring decisions.

2. **To design a personalized job recommendation system by utilizing collaborative filtering, content-based filtering, and hybrid recommendation algorithms, ensuring accurate matches based on user preferences and qualifications** :- The Job Recommendation System aims to develop a personalized job recommendation system using collaborative filtering, content-based filtering, and hybrid recommendation algorithms to enhance job matching accuracy. By analyzing user preferences, skills, experience and education , the system suggests the most relevant job opportunities. The hybrid approach ensures better adaptability and precision, improving the job search experience for candidates.
3. **To develop a market rate analysis tool using predictive analytics and time-series forecasting algorithms providing real-time insights into salary trends and benchmarks :-** The Job Recommendation System aims to develop a market rate analysis tool using predictive analytics and time-series forecasting algorithms to track salary trends and benchmarks. By analyzing historical salary data, industry trends, and market fluctuations, the tool provides real-time insights into compensation expectations for various roles. This enables job seekers, employers, and HR professionals to make data-driven salary decisions, ensuring competitive and fair compensation.
4. **To develop an AI-powered chatbot that leverages NLP for career guidance, job-related queries, and resume feedback :-** The AI-powered chatbot is designed to provide career guidance, job-related assistance, and resume feedback using Natural Language Processing (NLP). It understands user queries, offers personalized job recommendations, and provides insights into industry trends and required skills. With real-time interaction, the chatbot enhances job search efficiency by answering FAQs, refining resumes, and suggesting relevant job opportunities.

1.4 Scope:

The Job Recommendation and Resume Parsing System encompasses a broad range of functionalities designed to streamline the job search and recruitment process through advanced AI technologies. It efficiently parses and categorizes resume data using Natural Language Processing (NLP), ensuring accurate extraction of skills, experience, education, and certifications from unstructured resumes.

The scope extends to personalized job recommendations using collaborative filtering, content-based filtering, and hybrid recommendation algorithms to ensure that candidates receive job suggestions aligned with their qualifications, preferences, and career goals. By integrating Machine Learning (ML) algorithms, the system continuously improves the accuracy of its job matching, adapting to market trends and user behaviour.

With an intuitive interface and real-time job market insights, the system helps job seekers make informed career decisions, while employers can identify and acquire top talent effortlessly. By replacing traditional keyword-based job search and manual resume screening with data-driven, automated processes, the system enhances the overall job-seeking and hiring experience, making it faster, smarter, and more efficient for both job seekers and recruiters.

Chapter 2

Literature Review

Automated resume parsing and ranking using NLP has gained significant attention in recruitment. Sai Snehith K, Meena Belwal, Nithish Sagar Reddy, and Thatavarthi Giri Sougandh (2023) explored how Named Entity Recognition (NER) and Part-of-Speech (POS) tagging enhance resume parsing accuracy by efficiently extracting key details such as skills, experience, and education. Similarly, Gupta and Sharma (2019) studied semantic matching techniques using Word2Vec and BERT, demonstrating how these NLP models improve job-candidate compatibility by aligning resumes with job descriptions more effectively. Additionally, Kim and Park (2020) integrated Deep Learning models such as Bidirectional LSTMs and Transformers to improve resume ranking, ensuring a more accurate and efficient shortlisting process. These advancements contribute to a bias-free, data-driven hiring approach, reducing manual screening efforts while improving recruitment accuracy and efficiency. [1].

Job recommendation systems using collaborative filtering have gained significant attention in recent years. Wang, L., and Chen, Y. (2021) highlighted how user-based and item-based filtering improve job matching by analyzing user interactions. Lee, J., and Kim, H. (2019) explored hybrid models combining content- based and collaborative filtering for better recommendation precision. Zhang, X., and Zhao, R. (2020) demonstrated that deep learning models like Autoencoders and Neural Collaborative Filtering (NCF) enhance scalability and personalization. The proposed system builds on these advancements by integrating user behaviour analysis, deep learning, and adaptive job recommendations for a more efficient job search experience[2].

In the paper titled "A Machine Learning-Based Job Forecasting and Trends Analysis System to Predict Future Job Markets Using Historical Data," the authors explore the role of machine learning in analyzing job market trends. The research focuses on leveraging historical employment data and predictive analytics to identify emerging job opportunities and industry demands. The study highlights effectiveness of algorithms such as time-series forecasting, regression models, and neural networks in predicting future job trends. This approach promotes data-driven decision-making, enabling individuals to adapt to evolving job market conditions[3].

Chapter 3

Proposed System

The proposed system for a Job Recommendation and Resume Parsing with NLP and Chatbot project aims to provide a comprehensive and user-friendly solution for searching job efficiently. The proposed system aims to provide users with a comprehensive tool to enhance the skill and a better place to work, gain insights of the market trend for job.

3.1 Features and Functionality:

A job recommendation system project includes a range of features and functionalities designed to help individuals for searching job effectively. Here are common features and functionalities:

1. **Resume Parsing:-** The system analyzes and extracts critical details from resumes, including skills, experience, education, certifications, and job preferences. The NLP-based resume parser extracts and structures key information from resumes. Recruiters benefit from an automated filtering mechanism, saving time and effort in shortlisting candidates

2. **Job Recommendation :-** Job seekers receive tailored job recommendations based on their career history, preferences, and skill set. The system continuously learns and adapts to provide improved recommendations over time. Users can set career goals and receive job suggestions that align with their aspirations.

3. **Job Market Trend Analysis :-**Industry Insights & Future Job Demands: The system analyzes job market trends using historical and real-time data, identifying high-demand skills and emerging career opportunities. It helps job seekers align their skillsets with market needs, improving job readiness.

4. **AI Chatbot :-** A conversational chatbot assists job seekers by providing career advice, resume enhancement tips, and answering job-related queries. The chatbot serves as a virtual career assistant, available to provide job-related guidance.

Chapter 4

Requirements Analysis

The success of the Job Recommendation and Resume Parsing system hinges on its comprehensive software requirements, ensuring seamless functionality and an intuitive user experience. This AI-driven system integrates resume parsing, job recommendation, and chatbot assistance to enhance recruitment and job search processes. The platform leverages Natural Language Processing (NLP) for resume extraction, machine learning algorithms for job matching, and chatbots for career guidance. To achieve this, the system utilizes MySQL for backend data management and AI models for intelligent recommendations.

Here's how these software requirements contribute to the system's success:

Resume Parsing :

Automated Resume Analysis: The system extracts structured information from resumes using NLP techniques like Named Entity Recognition (NER) and Part-of-Speech (POS) tagging. The extracted details include skills, experience, education, and certifications, which are stored in a MySQL database for further processing.

Core Features:

- Upload and parse resumes in multiple formats (PDF, DOCX, etc.).
- Extract and classify key details (name, contact info, skills, experience, education,).
- Store parsed data in a structured database for efficient retrieval.

Job Recommendation System :

Intelligent Job Matching: The system employs machine learning and collaborative filtering to recommend job listings based on skills, experience, and preferences. It matches candidate profiles with job descriptions using semantic similarity techniques like TF-IDF.

Core Features:

- Personalized job recommendations based on extracted resume data.

- Semantic matching of resumes with job descriptions.
- Ranking of job opportunities based on candidate suitability.

Job Market Trend Analysis :

Industry Insights & Future Job Demands: The system analyzes job market trends using historical and real- time data, identifying high-demand skills and emerging career opportunities. It helps job seekers align their skillsets with market needs, improving job readiness.

Core Features:

- Tracks industry-specific job demand using historical and real-time data
- Identifies trending skills, certifications, and career paths.
- Provides market insights for informed career decisions.

Chatbot Assistance :

AI-Powered Career Guidance: A chatbot assists users by answering queries, suggesting job opportunities, and providing resume feedback. It uses Natural Language Understanding (NLU) to analyze user input and deliver relevant responses.

Core Features:

- Interactive chatbot for job search and resume feedback.
- Real-time assistance on job applications and career advice.
- Integration with job databases for instant job suggestions.

Chapter 5

Project Design

The design of the Job Recommendation and Resume Parsing System focuses on providing users with an intelligent and seamless job search experience. The system integrates core features such as resume parsing, AI-driven job recommendations, chatbot assistance, and market trend analysis to enhance career planning.

Using Natural Language Processing (NLP), the system extracts key details from resumes, while machine learning algorithms match candidates with the most relevant job opportunities. The chatbot offers real- time guidance, resume feedback, and job insights, ensuring an interactive and user-friendly experience.

The intuitive interface allows for smooth navigation, enabling users to upload resumes, explore job recommendations, and gain industry insights effortlessly. The system architecture ensures efficient data processing, providing users with accurate recommendations, career trends, and instant support for their job search.

5.1 Use Case Diagram

The use case diagram for Job Recommendation and Resume Parsing with NLP and Chatbot illustrates the user’s interaction with key functionalities: Resume Parsing, Job Recommendation, Market Trend Analysis, and Chatbot. This diagram visually represents how users can interact with these features, ensuring a personalized and efficient job search experience.This description outlines the design principles and key features of Job Recommendation and Resume Parsing with NLP and Chatbot, as shown in Figure (5.1.1).

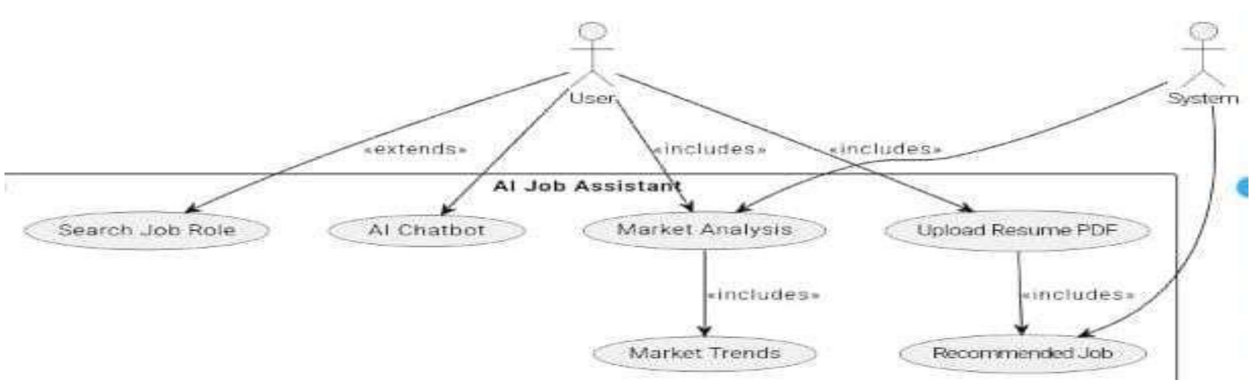

Figure 5.1.1 : Use case diagram

5.2 DFD (Data Flow Diagram)

The Data Flow Diagram (DFD) for the Job Recommendation and Resume Parsing with NLP and Chatbot platform illustrates the systematic movement of data between users, system processes, external systems, and data stores. Users interact with the platform by uploading resumes, searching for job recommendations, tracking job applications, and engaging with the chatbot for career guidance. These inputs are processed by the system to generate personalized job matches, resume insights, and real-time assistance, as shown in Figure 5.2.1.

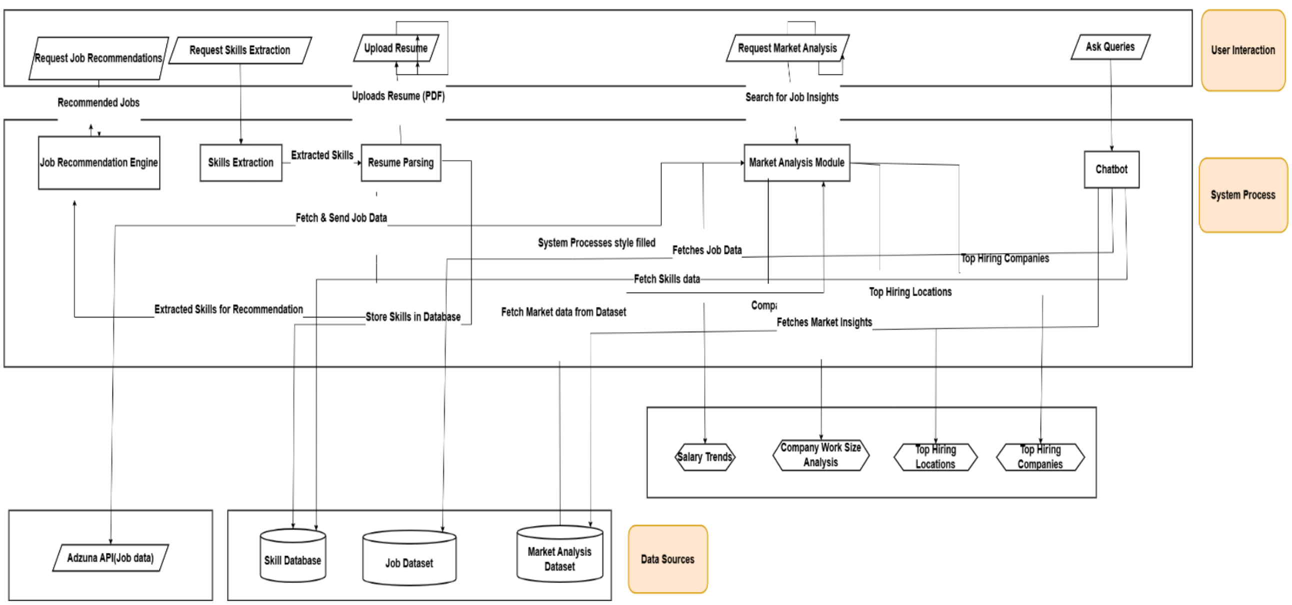

Figure 5.2.1: DFD - Data Flow Diagram

At the core of the system, multiple processes collaborate to enhance the job search experience. The Resume Parsing process extracts relevant details such as skills, experience, education, and qualifications from uploaded resumes, storing the structured data in the Resume Dataset (MySQL). This parsed data plays a crucial role in generating accurate job recommendations.

The Job Matching & Recommendation process compares user profiles against job listings using Natural Language Processing (NLP) and Machine Learning (ML) algorithms. It retrieves relevant job postings from the Job Listings Dataset, ensuring that users receive tailored job suggestions based on their qualifications and preferences.

The Chatbot Assistance process provides real-time support, answering user queries related to resume improvements, interview tips, job market trends, and application guidance. The chatbot accesses data from the Career Guidance Dataset, ensuring that users receive accurate and insightful responses.

The Skill Gap Analysis process identifies missing skills in a user's resume by comparing it with job requirements. The system then suggests relevant upskilling resources, courses, or certifications, storing recommendations in the Skill Development Dataset.

Data flows seamlessly between these stores and system processes, ensuring accurate and timely job recommendations, resume insights, and user assistance. Finally, the outputs from these processes are delivered to users in the form of personalized job matches, skill improvement suggestions, and real- time chatbot guidance. This structured flow ensures a seamless, AI-powered job search experience, helping users secure better career opportunities efficiently

5.3 System Architecture

This diagram represents a job recommendation and resume parsing model that integrates resume extraction, job matching, chatbot assistance, and job market trend analysis. Below is a breakdown of the various components and their functions as illustrated in Figure (5.3.1).

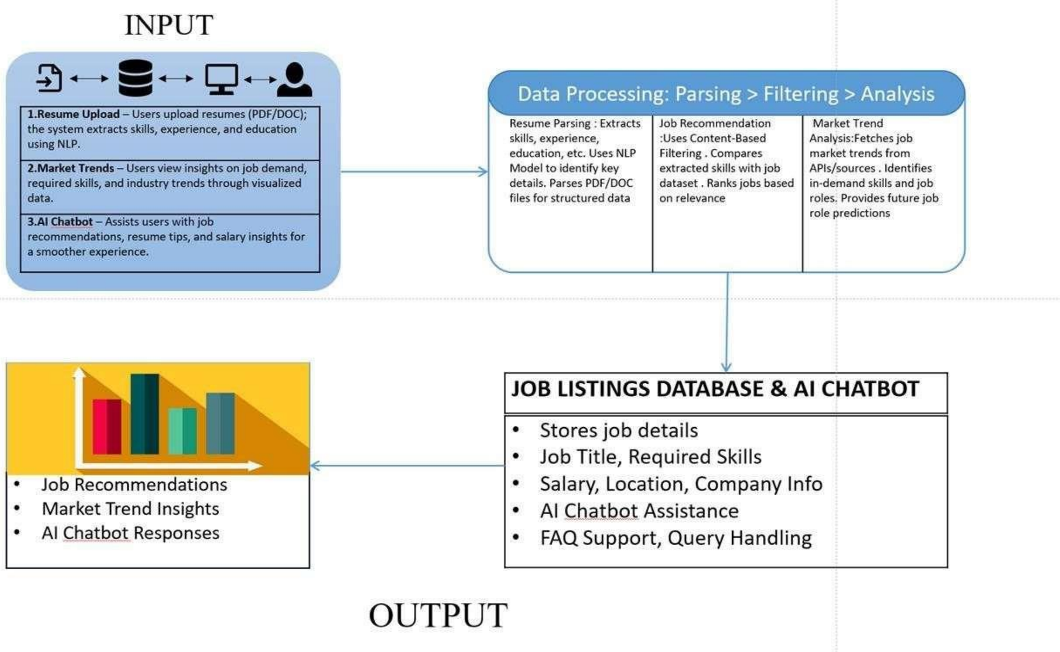

Figure 5.3.1 : System Architecture

INPUT Section:

The input consists of three key data sources related to job applications and career insights:

- Resume Upload: Users upload resumes in PDF/DOC format. The system extracts skills, experience, education, and certifications using NLP techniques.
- Market Trends: Users view job demand insights, required skills, and industry trends based on real- time data.
- AI Chatbot: Assists users with job recommendations, resume improvement tips, and salary insights for a better job search experience.

Data Processing Section:

The input data undergoes cleaning, parsing, and intelligent analysis, applying the following techniques:

Resume Parsing: Uses NLP models to extract structured information from resumes, including job titles, skills, and work experience.

Job Recommendation: Uses content-based filtering and semantic similarity techniques (BERT, TF-IDF) to match candidate skills with job descriptions and rank jobs based on relevance.

Market Trend Analysis: Fetches job market trends from APIs and other sources. It identifies high-demand skills, emerging job roles, and future career predictions.

MODEL Section:

The system integrates AI-driven models for job recommendations and career insights: Resume Parsing Model extracts key resume details using NER and POS tagging. Job Matching Model uses content-based filtering and semantic similarity for personalized job recommendations. Chatbot & Market Trend Models provide real-time career advice, salary insights, and industry trend analysis to enhance job search decisions.

OUTPUT Section:

The system provides personalized job recommendations, ranking openings by relevance. Market trend insights offer visual reports on emerging skills and job demands. The AI chatbot and job listings database assist with real-time job guidance, resume feedback, and detailed job information.

5.4 Activity Diagram

The diagram represents a comprehensive workflow for an AI-powered job recommendation system, which integrates resume parsing, job market analysis, intelligent job recommendations, and an AI chatbot assistant to enhance user experience and job-matching efficiency as shown in Figure (5.4.1).

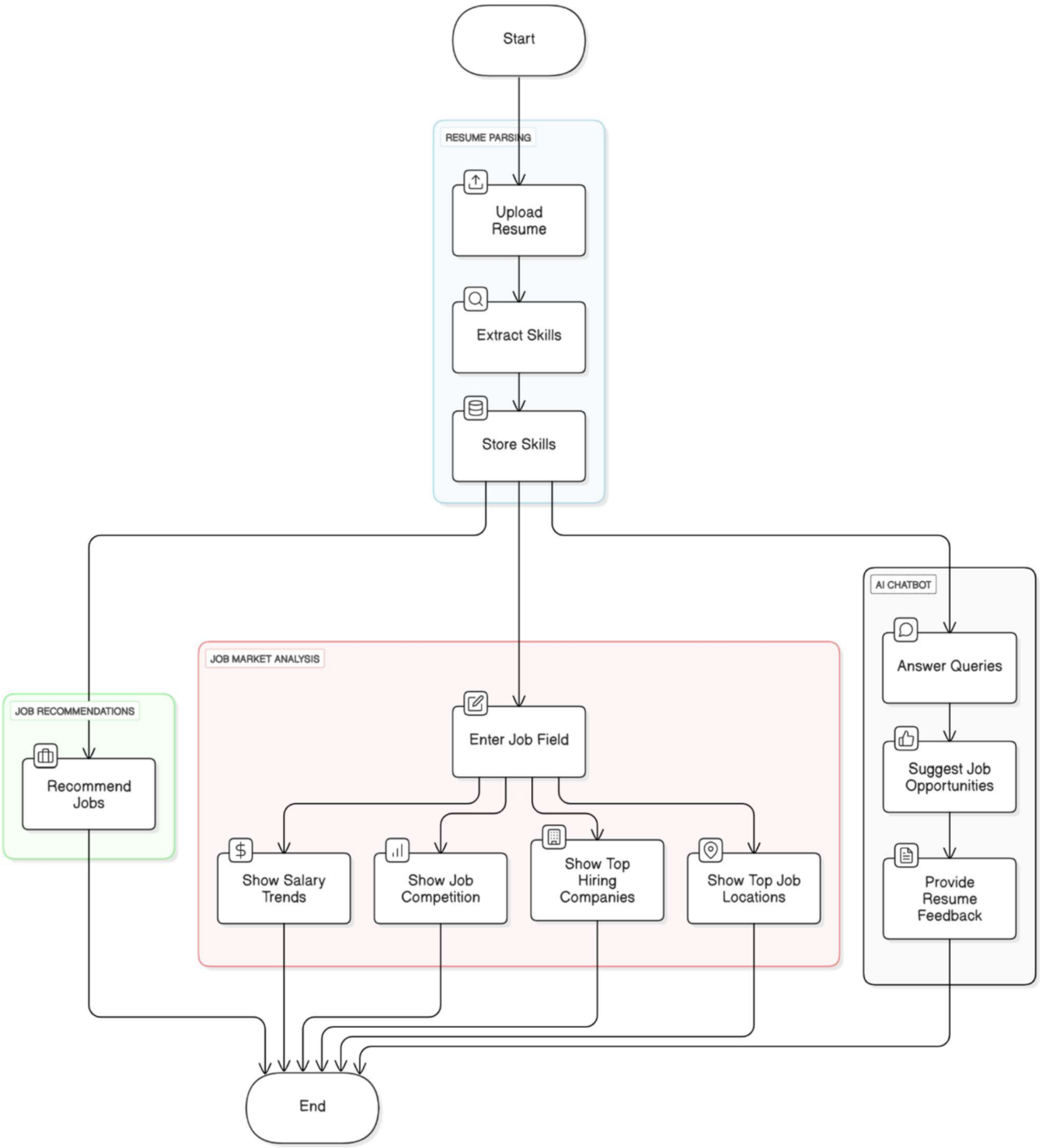

Figure 5.4.1 : Activity Diagram

1. Start:

   The system begins when a user initiates the process to find job opportunities by uploading their resume.

2. Resume Parsing Module:

   This module is responsible for extracting and managing user information, particularly their skills:

- Upload Resume: Users upload their resumes in supported formats (PDF, DOCX, etc.).
- Extract Skills: The system uses Natural Language Processing (NLP) techniques to parse the resume content and extract relevant skills, qualifications, and work experiences.
- Store Skills: The extracted information is stored in a structured format within a database for further use in job matching and analysis.

3. Job Market Analysis Module :

   This segment helps users explore the current job market landscape based on their input: Enter Job Field: Users input their desired job domain (e.g., Data Science, Web Development).

   The system then provides insights into:

   - Show Salary Trends: Displays current salary ranges and trends for the entered job field.
   - Show Job Competition: Indicates the level of competition or demand for jobs in that field.
   - Show Top Hiring Companies: Lists leading companies currently hiring for the specified domain.
   - Show Top Job Locations: Identifies geographical regions with the most job opportunities for the selected field.

   This helps job seekers make informed decisions about job selection and skill development.

4. Job Recommendation Engine :

   Based on the parsed skills and market analysis:

   Recommend Jobs: The system suggests suitable job openings tailored to the user's skill set and preferences using machine learning models such as collaborative filtering or content-based filtering.

5. AI Chatbot Assistant :

An AI-powered chatbot enhances user interaction and offers real-time assistance:

- Answer Queries: Responds to user questions regarding job searches, system usage, and other general inquiries.
- Suggest Job Opportunities: Recommends jobs based on user interaction and stored skill data.
- Provide Resume Feedback: Offers suggestions to improve resume quality, increasing the chances of being shortlisted.

6. End :

After completing all the steps—parsing, analysis, recommendations, and interactions—the process concludes, with the user equipped with personalized job opportunities and relevant job market insights.

5.5 Implementation

1. Upload Resume Page :

User can upload the resume on clicking the choose file as shown in Figure (5.5.1) .

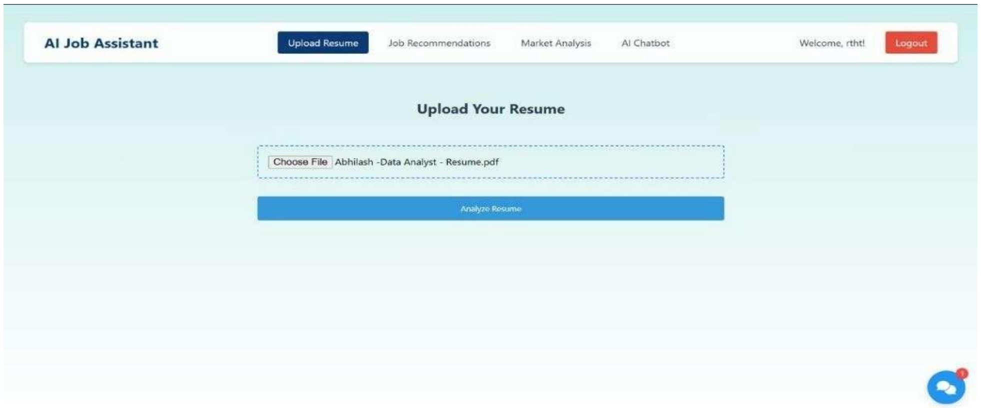

Figure 5.5.1: Upload Resume Page

2. Parse Resume Page :

Once user has uploaded the resume , using NLP techniques like Named Entity Recognition (NER) skills are extracted and on clicking the submit button skills are stored in database, as shown in Figure(5.5.2).

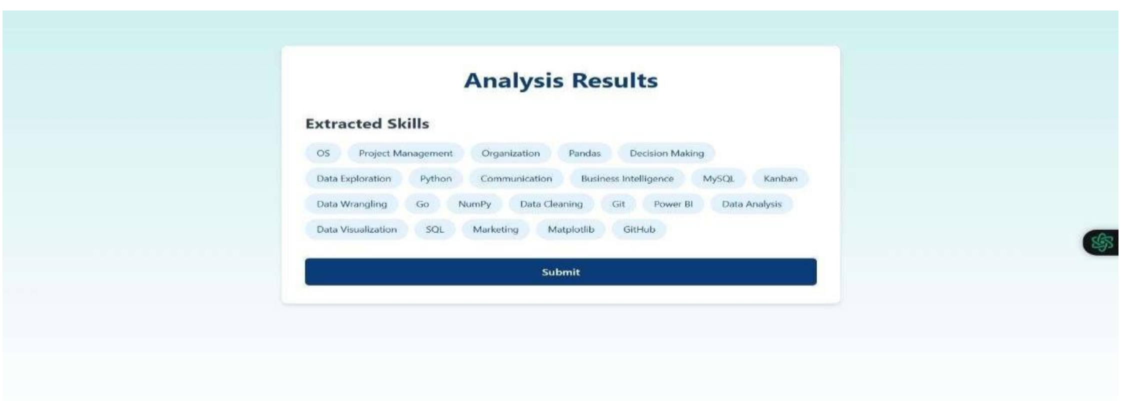

Figure 5.5.2: Parse Resume Page

3. Job Recommendations Page:

Depending on the skills extracted from the uploaded resume some top jobs are recommended , as shown in Figure(5.5.3).

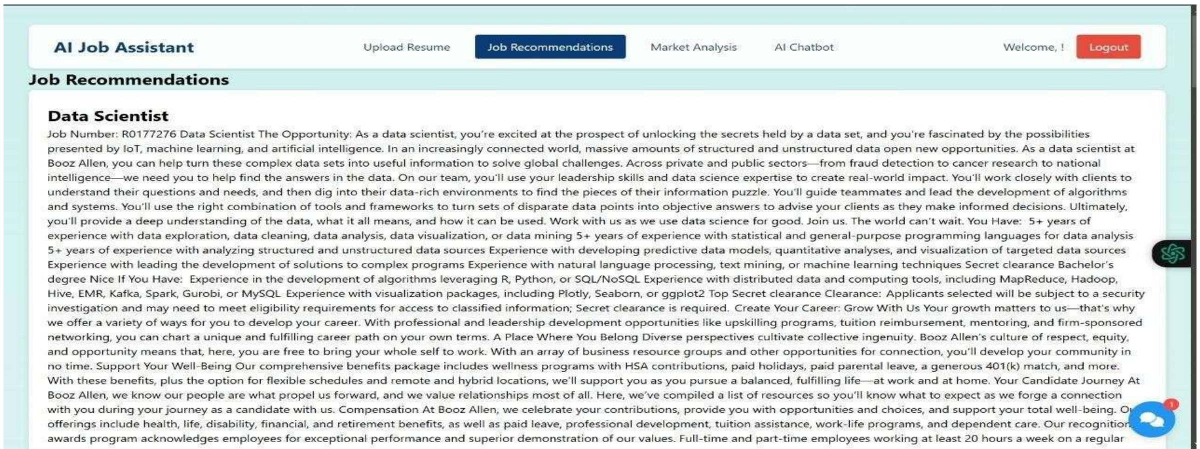

Figure 5.5.3 : Job Recommendations Page

4. Job Market Analysis Page:

User enter the job, related to any field in which he/she want to know the market trends. It will show the Salary Trends, Job Competition Analysis, Top Hiring Companies and Top Job Location, as shown in Figure (5.5.4) and Figure (5.5.5).

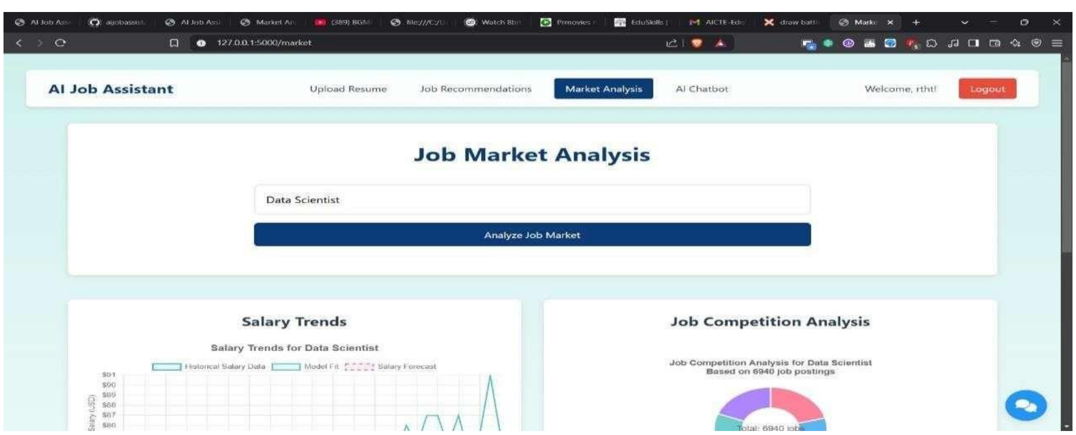

Figure 5.5.4 : Job Market Analysis Page

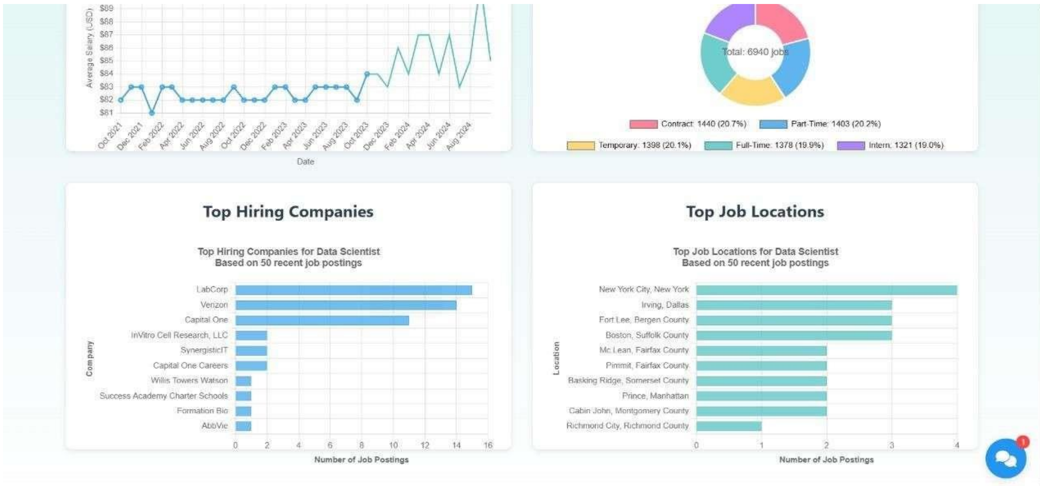

Figure 5.5.5 : Job Market Analysis Page

5. AI Chatbot :

AI Chatbot assists users by answering queries, suggesting job opportunities, and providing resume feedback, as shown in Figure (5.5.6).

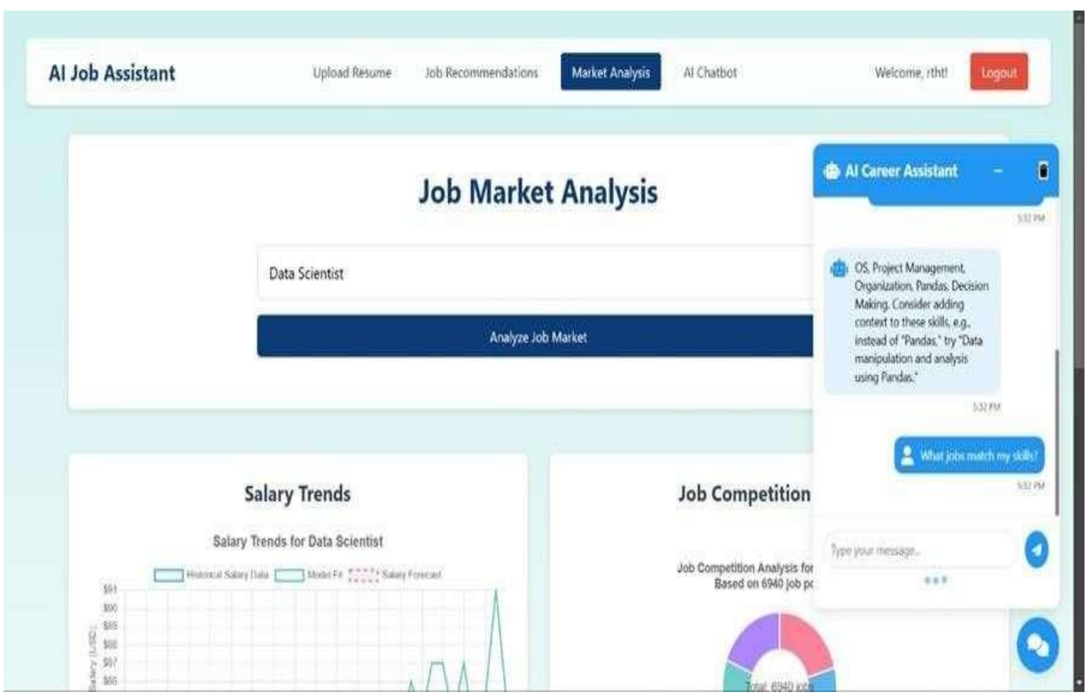

Figure 5.5.6: AI Chatbot

Chapter 6

Technical Specifications

In the development of the Job Recommendation and Resume Parsing with NLP and Chatbot platform, a variety of technologies work together to build a robust system for recommendation, and analysis in searching job. Here's how each component is utilized:

Frontend Technologies:

- **HTML (version 5.0):** HTML structures the web pages of the Job Recommendation and Resume Parsing with NLP and Chatbot platform, forming the backbone of the user interface. It ensures that the content is well-organized, allowing users to easily navigate through different features.
- **CSS (version 3.0):** CSS is employed for styling the platform, making it visually appealing and responsive across devices. CSS ensures that the platform has a clean, modern, and user-friendly interface, which is important for finding a better job.
- **JavaScript (version ES6):** JavaScript is used to add interactivity to the Job Recommendation and Resume Parsing with NLP and Chatbot platform. From Setting Resume Parsing to AI Chatbot, JavaScript ensures that the frontend is interactive and responds quickly to user actions.

Backend Technologies:

- **Python (version 3.12.2):** Python handles the backend logic of Job Recommendation and Resume Parsing with NLP and Chatbot, including processing user inputs, managing requests from the frontend, and implementing algorithms. Python is used to connect to datasets, apply machine learning models for recommendation, and display relevant data like recommended job, market trend.
- **Flask (version 3.0.3):** Flask is used as the web framework to manage routing and

handle server-side logic. It ensures efficient handling of requests, such as processing user registrations and recommendation requests. Flask is also used to integrate with external APIs for more accuracy

Database Management:

MySQL (version 8.1.0): MySQL serves as the database management system for Job Recommendation and Resume Parsing with NLP and Chatbot. It is used to store large sets of data, including user profiles, parsed data. MySQL ensures that the platform can efficiently store, retrieve, and manage vast amounts of data necessary for providing reliable predictions and recommendations.

Data and Machine Learning:

- **Job Description Datasets:** The system processes extensive datasets containing detailed job descriptions, covering job titles, required skills, qualifications, responsibilities, and industry trends. By leveraging machine learning models, the system analyzes these datasets to generate tailored job recommendations, identify skill gaps, and provide career guidance, helping users make informed employment decisions(it includes Columns =23 , Rows = 7000).
- **Job Listing Datasets:** The system utilizes various datasets containing historical and real-time job market data, including job titles, required skills, salaries, and industry trends. These datasets are analyzed using machine learning models to provide users with valuable insights, such as personalized job recommendations, market demand analysis, and career growth opportunities(it includes Columns =25 , Rows = 1000).
- **Skill Datasets:** The system utilizes various datasets containing historical job market data, including users skills,and industry trends. These datasets are analyzed using machine learning models to provide users with personalized job recommendations, skill analysis, and career growth opportunities (it includes Columns =2, Rows = 900).

External Integrations:

APIs for Job Market Analysis : The Job Recommendation system connects to external services to fetch real-time job market data, which plays crucial role in providing accurate job insights. These integrations allow the platform to analyze industry trends, demand for specific skills, top hiring companies, top job locations, helping job seekers make informed career decisions. By leveraging these datasets, machine learning algorithms user can predict future job market trends, identify emerging skill requirements, and offer personalized job recommendations, assisting users in aligning their career paths with evolving industry demands.

Chapter 7

Project Scheduling

The Job Recommendation and Resume Parsing System project began with the Project Conception and Initiation phase, which was successfully completed by 4/4/2025. During the first week, the group—Shuban Devanpelli, Dhanraj Bacche, Yash Kamble, and Rishi Bije—finalized the project topic and outlined the scope and objectives of the mini-project. From 8/1/2025 to 11/1/2025, the team identified the core functionalities necessary for implementation. Subsequently, they collaboratively refined the project concept with the help of a paper prototype between 15/1/2025 and 18/1/2025, ensuring every member contributed to visualizing the idea.

The Graphical User Interface (GUI) design began on 18/1/2025 and was completed by 11/2/2025, resulting in a user-friendly interface for the application. The initial development phase culminated in Presentation I on 11/2/2025, where the team showcased the planned architecture, designs, and system flow, as represented in Figure (7.1).

Moving into the Project Design and Implementation phase, technical development took precedence. The team began with Database Design from 15/2/2025 to 23/2/2025, laying the groundwork for storing parsed resume data and job listings. Following this, from 24/2/2025 to 5/3/2025, the group established Database Connectivity for all modules, enabling seamless data exchange between system components.

From 8/3/2025 to 21/3/2025, the focus shifted to module integration and report writing. During this time, the team ensured the chatbot, job recommender, and resume parser were all fully functional and well-integrated. The project culminated in Presentation II on 2/4/2025, where the team demonstrated the end-to-end working system—including NLP-based parsing, intelligent recommendations, and interactive chatbot functionality—thus marking the successful completion of the project.

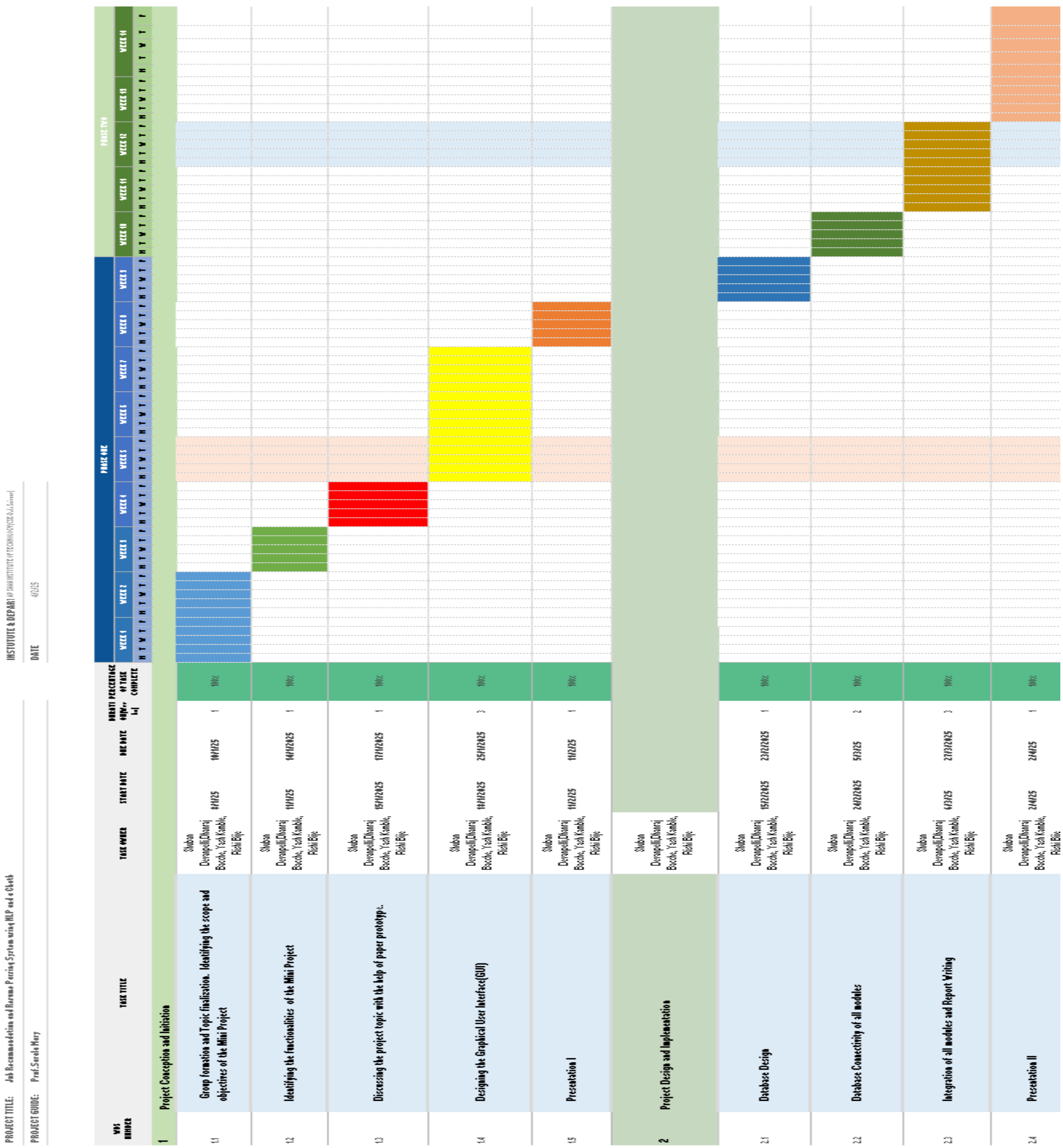

Figure 7.1: Gantt Chart

Chapter 8

Result

The Job Recommendation and Resume Parsing system has been developed with two main model-based features: Resume Parsing (for candidate information extraction) and Job Recommendation (for intelligent job matching). Both models were evaluated based on performance metrics like Precision, Recall, F1-score, and accuracy using real-world recruitment datasets. Below is a detailed summary of the evaluation results.

8.1 Resume Parsing

The Resume Parsing feature employs Named Entity Recognition (NER) and Part-of-Speech (POS) tagging to extract key details such as skills, experience, education, and certifications from resumes. The model was trained on a dataset containing structured and unstructured resumes.

This bar chart visualizes a comparison between the actual skills listed by a user and the extracted skills identified by the system from their resume. The x-axis represents the skill types, while the y-axis shows the count of skills. From the chart, it's evident that the number of extracted skills slightly exceeds the actual skills. This indicates that the skill extraction algorithm was effective in identifying both explicitly stated and possibly implied skills, making it useful for enhancing the accuracy of job recommendations., as shown in Figure 8.1.

Visualization of Model Performance:

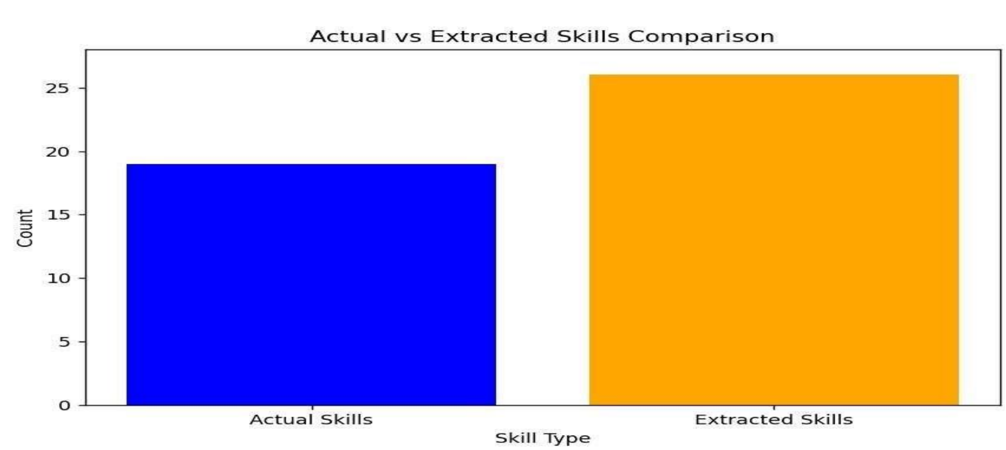

Figure 8.1: Actual vs Extracted Resume Details

8.2 Job Recommendation

The Job Recommendation feature uses a hybrid collaborative filtering model that combines content-based filtering and collaborative filtering to suggest relevant job opportunities based on a candidate's resume data, preferences, and job market trends. The model was trained on job descriptions and user application history and evaluated using accuracy, Mean Average Precision (MAP), and Recall@K.

Model Performance:

Mean Average Precision (MAP): 85.7%

Recall@10: 88.3%

Accuracy: 96.5%

These results suggest that the model provides highly relevant job recommendations, ensuring strong alignment between job descriptions and candidate profiles. However, periodic model retraining is required to accommodate evolving job market trends and new user preferences, as illustrated in Figure 8.2.

Visualization of Model Performance:

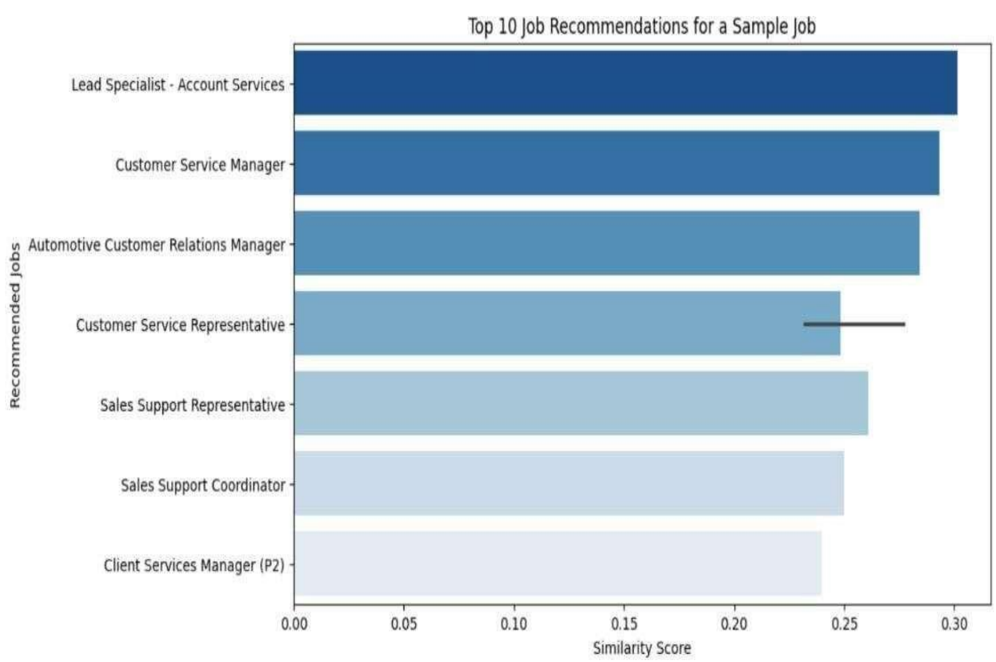

Figure 8.2: Job Recommendation Precision and Recall Analysis

Chapter 9

Conclusion

The Job Recommendation and Resume Parsing system with NLP and Chatbot is an advanced recruitment solution that enhances job matching efficiency and candidate evaluation. By leveraging Natural Language Processing, AI-driven chatbots, and machine learning algorithms, the system streamlines resume parsing, improves job recommendations, and provides personalized career guidance. With its intelligent automation and user-friendly interface, it not only simplifies the hiring process for recruiters but also empowers job seekers with real-time insights and tailored job opportunities. The project team is dedicated to continuous innovation and optimization, ensuring the system remains adaptable to evolving industry needs. As a future-ready solution, the system aims to revolutionize the recruitment landscape, making job searching and hiring smarter, faster, and more effective.

Chapter 10

Future Scope

The future scope of the Job Recommendation and Resume Parsing with NLP and Chatbot could include several enhancements and additional features to further improve the user experience and provide more comprehensive terms related to job searching. Here are some potential future enhancements:

Enhanced AI-Powered Matching : Future advancements in AI and NLP can improve job-candidate matching by integrating deep learning models like transformers and reinforcement learning to provide more accurate recommendations.

Real-Time Resume Optimization : AI-driven resume parsing can evolve to offer instant suggestions for resume improvements based on industry trends, recruiter preferences, and ATS (Applicant Tracking System) requirements.

Voice and Conversational AI Integration : The chatbot can be enhanced with voice recognition and conversational AI to provide a more interactive and personalized job search experience, assisting users in real-time with job recommendations and interview preparation.

Cross-Platform Job Aggregation : The system can be expanded to integrate with multiple job portals, LinkedIn, and freelancing platforms to provide a centralized job search experience for users.

Personalized Career Pathway Recommendations : Leveraging AI and big data analytics, the system can suggest skill development courses, certifications, and career growth strategies based on user profiles and industry demands.

Bias Reduction and Fair Hiring Practices : Future developments can focus on reducing hiring biases by implementing explainable AI models that ensure fair candidate screening and ranking.

Integration with HR and ATS Systems : The system can be seamlessly integrated with enterprise ATS platforms, enabling recruiters to automate candidate selection workflows.

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